# IoT-CAD

**:** IoT-CAD: Context-Aware Adaptive Anomaly Detection in IoT Systems Through Sensor Association

**:** Rozhin Yasaei, Felix Hernandez Mohammad Abdullah Al Faruque

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IoT-CAD

LSTM encoder-decoder

multivariate Gaussian

consensus

62 13

8 2.3

IoT-CAD precision 92 recall 56

F0.5 81 F1 70

0.532 fog node

1

Environmental Anomaly EA

Sensing Device Anomaly SDA

IoT

2 IoT-CAD

binary fingerprint

1

0

fingerprint Hamming distance

clustering

LSTM encoder-decoder LSTM

reconstruction error error

error

alpha

unsupervised validation

alpha F-score

consensus

EA

SDA

IoT-CAD fog computing

fog nodes

predictive model

## 3

62 13

IEEE 802.15.4 software-defined radio Raspberry Pi gateway Intel Core i5-6300HQ RAM 8 GB fog node

8 2.3 Table 2 IoT-CAD precision 92 LSTM baseline 64 ConvLSTM baseline 51 One-Class SVM precision 89 IoT-CAD F0.5 F1 81 70 recall IoT-CAD 56 ConvLSTM recall 95 trade-off IoT-CAD false positive

| | Precision | Recall | F0.5 | F1 |
|---|---|---|---|---|
| **IoT-CAD** | **92%** | **56%** | **81%** | **70%** |
| LSTM baseline | 64% | 44% | 58% | 52% |
| ConvLSTM baseline | 51% | 95% | 56% | 66% |
| One-Class SVM | 89% | 25% | 60% | 39% |

1 Table 2

predictive model 2,472.20 41.2 0.532

real time noise Gaussian, pink uniform precision noise IoT-CAD precision

EA

## 4 Contribution

Contribution sensor association
binary fingerprint

Contribution consensus anomaly detector

EA SDA

SDA

Contribution
precision

## 5

machine learning
sensor network
false alarm
testbed 62
precision, recall, F-score, noise

fog computing
latency AI
AI
LSTM
sensor integrity, concept drift
anomaly edge AI, industrial IoT digital twin

## 6

recall 56
56
precision 92 92
false negative
recall threshold

anomaly

8

consensus

threat model

consensus

EA

model poisoning anomaly

## 7

sensor network, embedded communication, fog computing, time-series machine learning cybersecurity

5 8

AI LSTM

AI

sensor fault

LSTM transformer

latency, edge device

false negative false positive

coordinated attack


[1] R. Yasaei, F. Hernandez, and M. A. Al Faruque, "IoT-CAD: Context-Aware Adaptive Anomaly Detection in IoT Systems Through Sensor Association," in *2020 IEEE/ACM International Conference on Computer-Aided Design (ICCAD)*, 2020, pp. 1–9, doi: 10.1145/3 400 302.3415672.

[2] IEEE, "IEEE 802.15.4 Standard for Low-Rate Wireless Networks," IEEE Standards Association. https://standards.ieee.org/standard/802_15_4-2020.html